

Solutions for TMA 02 2025J

Part B Formative

Solution to Question 8 – 9 marks

- (a) (i) We proceed by contradiction.

Assume that n is prime. Then $\sigma(n) = \sigma(p) = 1 + p$ by

Proposition 1.7, HB pg 42 or from the definition of σ .

Therefore $4n + \sigma(n) = 4p + 1 + p = 5p + 1$.

Therefore if $4n + \sigma(n)$ is divisible by 5 and n is a prime p we have that $5p + 1$ is divisible by 5 and this is impossible. Therefore n cannot be prime.

- (ii) Now let $n = p^2$. By **Proposition 1.7, HB pg 42** we have that $\sigma(p^2) = 1 + p + p^2$. Therefore $4n + \sigma(n) = 4p^2 + 1 + p + p^2 = 5p^2 + 1 + p$.

This means that if $4n + \sigma(n)$ is divisible by 5 then we must have $5p^2 + 1 + p$ divisible by 5 and so $1 + p$ is divisible by 5. In other words, p is congruent to 4 (mod 5).

- (iii) We know that n cannot be prime and that $p \equiv 4 \pmod{4}$ if $n = p^2$, where p is prime. So we can check non primes, $n = 4, 6, 8, 9, 10, 12$ and 14, and find that $\sigma(14) = 24$ and $4 \times 14 + 24 = 80$; or we can check numbers that are equal to a prime squared where that prime is 4 mod 5. The first prime that is also 4 mod 5 is 19. Calculating gives $4 \times 19^2 + \sigma(19^2) = 4 \times 19^2 + 1 + 19 + 19^2 = 1825$. So possible answers are 14 or 19^2 . There are others (such as 16, 33, 34, 36, 40, 44, 74, 78, 81, 84, 93, 94, 95).

- (b) Let p and q be distinct primes.

Then $pq + \sigma(pq) = pq + 1 + p + q + pq = 2pq + 1 + p + q$

Proposition 1.7, HB pg 42.

So if $pq + \sigma(pq)$ is even then we must have $1 + p + q$ even and so $p + q$ must be odd. Therefore the pairs are $(2, q)$ for any odd prime q .

- (c) Let $m = p_1^{m_1} \cdots p_r^{m_r}$. Then $\tau(m) = (m_1 + 1) \cdots (m_r + 1)$

Proposition 2.3 of Chapter 2, HB pg 17. Therefore if $\tau(m) | m$ we must have $(m_1 + 1) \cdots (m_r + 1) | m$. But since m is odd, we must have $m_i + 1$ being odd for all i . Therefore m_i is even, say $m_i = 2k_i$. Then $m = p_1^{m_1} \cdots p_r^{m_r} = p_1^{2k_1} \cdots p_r^{2k_r} = (p_1^{k_1} \cdots p_r^{k_r})^2$ as required.

Solution to Question 9 – 8 marks

We need to check the group axioms, G1, G2, G3 and G4

Definition 1.1 of Chapter 5, HB page 24.

Closed

If f and g are both continuous on $[0, 1]$ then so is $(f + g)$. Hence $(C[0, 1], +)$ is closed under $+$ (this is a standard result from, for example, the sum rule from M208).

Identity

Let $0_{C[0,1]}$ be the function defined by $0_{C[0,1]}(x) = 0$ for all $x \in [0, 1]$. Then for any $g \in C[0, 1]$ we have $(g + 0_{C[0,1]})(x) = g(x) + 0 = g(x)$ and $(0_{C[0,1]} + g)(x) = 0 + g(x) = g(x)$ and so $0_{C[0,1]}$ is the additive identity (and is continuous since it is a constant function).

Inverses

Let $f \in C[0, 1]$ be any element of $C[0, 1]$ and let $g : [0, 1] \rightarrow \mathbb{R}$ be defined by $g(x) = -f(x)$. Then g is continuous on $[0, 1]$ (by standard results from M208) and so $g \in C[0, 1]$. Since $(f + g)(x) = f(x) + (-f(x)) = f(x) - f(x) = 0$ for all x , g is the additive inverse of f .

Associative

For any f, g and h in $C[0, 1]$, we have

$$\begin{aligned} ((f + g) + h)(x) &= (f + g)(x) + h(x) = (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x)) \\ &\text{(by associativity in } \mathbb{R}) = (f + (g + h))(x). \end{aligned}$$

Solution to Question 10 – 9 marks

(a) $H = \langle r^2 \rangle = \{e, r^2, r^4, r^6, r^8, r^{10}\}.$

(b) The elements of H have the form r^{2k} for $k = 0, 1, \dots, 5$. We have

(i) $r^m r^{2k} r^{-m} = r^{2k}$, so $r^m H (r^m)^{-1} \subseteq H$ for $m = 0, 1, \dots, 11$, and

(ii) $r^m s r^{2k} (r^m s)^{-1} = r^m r^{-2k} s s^{-1} r^{-m} = r^{-2k} \in H$, so $r^m s H (r^m s)^{-1} \subseteq H$.

Therefore $gHg^{-1} \subseteq H$ for all $g \in D_{12}$, and therefore H is a normal subgroup of D_{12} by

Proposition 3.7(a) of Chapter 5, HB page 27.

Alternatively:

$$He = H = eH \text{ and}$$

$$rH = \{r, r^3, r^5, r^7, r^9, r^{11}\} = Hr.$$

Moreover, we have

$$sr^2 = r^{11}sr = r^{22}s = r^{10}s$$

and, similarly, $sr^{2k} = r^{12-2k}s$ for $k = 2, 3, 4, 5$, and so

$$sH = \{s, r^2s, r^4s, r^6s, r^8s, r^{10}s\} = Hs.$$

(Or: using $sr^{-i} = r^i s$ (HB p34) we have $sr^{2k} = r^{-2k}s = r^{12-2k}s$.)

The remaining six elements of D_{12} must form the fourth coset (by the proof of Lagrange's Theorem and

Proposition 3.7(b) of Chapter 5, HB page 27), and so

$$rsH = \{rs, r^3s, r^5s, r^7s, r^9s, r^{11}s\} = Hrs.$$

Therefore H is a normal subgroup of D_{12} by

Proposition 3.7(b) of Chapter 5, HB page 27.

(c) (i) The cosets are

i. H

ii. $rH = \{r, r^3, r^5, r^7, r^9, r^{11}\}$

iii. $sH = \{s, r^2s, r^4s, r^6s, r^8s, r^{10}s\}$

iv. $rsH = \{rs, r^3s, r^5s, r^7s, r^9s, r^{11}s\}$

The Cayley table is as follows.

	H	rH	sH	rsH
H	H	rH	sH	rsH
rH	rH	H	rsH	sH
sH	sH	rsH	H	rH
rsH	rsH	sH	rH	H

(ii) D_{12}/H is not cyclic, since there is no element of order 4, (or all non-identity elements have order 2).

Chapter 7, Exercise 1.4 and solution, pages 156 and 194–5

Chapter 5, Exercise 3.10 and solution, pages 37 and 74

Solution to Question 11 – 16 marks

- (a) $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_2 \times \mathbb{Z}_6$ and $\mathbb{Z}_{12} \cong \mathbb{Z}_3 \times \mathbb{Z}_4$

Corollary 2.12 of Chapter 6, HB page 32.

- (b)

Group	Order of group	Isomorphic groups	Orders of elements	Cyclic? Abelian?
Dic_6	24	None	1, 2, 3, 4, 6, 12	Non abelian
$\mathbb{Z}_3 \times \mathbb{Z}_4$	12	$\mathbb{Z}_{12}$	1, 2, 3, 4, 6, 12	Cyclic
$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3$	12	$\mathbb{Z}_2 \times \mathbb{Z}_6$	1, 2, 3, 6	Abelian, non cyclic

Chapter 6, Exercise 2.8 and solution, pages 111 and 142

Chapter 6, Exercise 2.9 and solution, pages 112 and 142

Chapter 6, page 112, Theorem 2.11

- (c) $\mathbb{Z}_3 \times \mathbb{Z}_4$ and Dic_6 .

Solution to Question 12 – 8 marks

- (a) (i) Here K is not a union of conjugacy classes of D_{10} , hence by

Proposition 3.12(a) of Chapter 5, HB page 28 it is not normal in D_{10} . Then D_{10} is not isomorphic to the direct product $H \times K$ (that is, it is not the internal direct product of H and K) by

Theorem 1.5 of Chapter 6, HB page 30.

- (ii) Since both H and K are unions of conjugacy classes, they are normal subgroups of D_{10} by

Proposition 3.12(b) of Chapter 5, HB page 28. Moreover, their intersection is $\{e\}$, and HK contains the generators r (because $r = r^6 r^5$) and s , and so $HK = D_{10}$. Hence $D_{10} \cong H \times K$ by

Theorem 1.5 of Chapter 6, HB page 30.

Chapter 6, Exercise 1.7 and solution, pages 96 and 138

- (b) Let p and q be distinct primes.

- (i) Since p^2 and q are coprime, $\mathbb{Z}_{p^2q}$ is isomorphic to $\mathbb{Z}_{p^2} \times \mathbb{Z}_q$ by

Corollary 2.12 of Chapter 6, HB page 32.

- (ii) By **Proposition 2.8 of Chapter 6, HB page 32**, the highest order of an element of $\mathbb{Z}_{p^2q}$ is p^2q . The maximum order of an element of $\mathbb{Z}_p \times \mathbb{Z}_p \times \mathbb{Z}_q$ is pq . Hence the two groups are not isomorphic by

Theorem 4.6 of Chapter 5, HB page 27.

Alternatively:

$\mathbb{Z}_p \times \mathbb{Z}_p$ is not cyclic by **Theorem 2.11 of Chapter 6, HB page 32**, hence $\mathbb{Z}_p \times \mathbb{Z}_p \times \mathbb{Z}_q$ has a non-cyclic subgroup, and it cannot be cyclic by

Theorem 2.3 of Chapter 6, HB page 31. Since $\mathbb{Z}_{p^2q}$ is cyclic, we have

$$\mathbb{Z}_p \times \mathbb{Z}_p \times \mathbb{Z}_q \not\cong \mathbb{Z}_{p^2q}.$$

Chapter 5, page 44, discussion following Exercise 4.9

Chapter 6, Exercise 2.9 and solution, pages 112 and 142